Project Phase 2: ER Model

CS4.301 Data and Applications

Deadline: 23:59, 30th October

1 The Task [40 Marks]

Design an ER diagram for your mini-world using the requirements document submitted in Project Phase 1. This ER diagram will serve as the foundation for your database design and implementation in future phases of the project.

Just like any ER diagram, we expect it to contain information about all entities (and their attributes) and the relationships they are in as per the requirements document.

2 Requirements [10 Marks]

Your ER diagram must be based on your Project Phase 1 requirements document and should accurately represent all specified components.

You are free to make **minor edits** to your mini-world (Please try to keep them to a minimum as the entire goal of the project is for you to scope out potential issues in the future while designing a database). **Mention these changes** in the submission document as well.

3 Tools

You are free to draw your ER diagram using any digital tool such as creately, draw.io, Lucidchart, or any viable alternatives.

Important: Hand-drawn diagrams will NOT be accepted. All submissions must use digital diagramming tools.

4 Guidelines [35 Marks]

We urge you to use the notation used by the **Textbook** (minor variations are acceptable considering the tool's limitations, please **clearly mention them** if any) to maintain clarity and standard representation.

Use (min-max) notation for cardinality and participation constraints.

Ensure that you spend some time on this as it's easy to miss some details with ER diagrams. Common mistakes include missing cardinality constraints, ambiguous participation constraints, and undifferentiated attributes (multivalued, primary key, etc). **Keep a lookout for these**.

5 Design Quality Analysis [15 Marks]

Analyze your ER diagram for potential data quality issues. For each issue you identify, explain how it might cause problems:

5.1 Redundancy Analysis

Identify any places where the same information might be stored multiple times. What problems could this cause?

5.2 Update Anomaly

If you need to change a piece of information, would you have to update it in multiple places? If yes, give a specific example from your design.

5.3 Insertion Anomaly

Are there situations where you cannot insert data because other related data doesn't exist yet? Describe one scenario.

5.4 Deletion Anomaly

If you delete certain data, would you unintentionally lose other important information? Give an example.

Note: No penalties for these patterns existing in your design. If your design naturally avoids certain patterns, simply note that.

6 Submission Instructions

Submit a single PDF named `<teamnumber>.pdf` (without the `<` and `>`). The PDF can consist of multiple pages if needed (e.g., for noting changes) but must contain the entire ER diagram on a single page.

Best of luck!